

Two-Phase PageRank with Travel-Time Self-Loops

A Complete Mathematical Reference
Including Eigenvector Equivalence and the Google Matrix
Salt Lake City Urban Traffic Prediction

Technical Documentation

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Abstract

We present a complete derivation, theoretical analysis, and empirical evaluation of the *Two-Phase PageRank with Travel-Time Self-Loops* algorithm, developed to predict macroscopic traffic congestion on the Salt Lake City road network (99,716 directed road links). Standard PageRank suffers from a fundamental physical defect: treating every link transition as instantaneous causes excessive probability-mass diffusion and a Mean Relative Error (MRE) of **0.7331** on the busiest 100 links.

We resolve this by injecting physical dwell time into the transition matrix via *self-loop weights* proportional to each link’s traversal time, controlled by a single global parameter μ . The network is split into two phases—active driving and local dispersal—forming a $2N \times 2N$ block transition matrix. With optimal parameters ($\mu = 20$, $\beta = 0.90$, $d = 0.80$) the Top-100 MRE reduces to **0.0317** (a **95.7%** improvement) while the overall MRE simultaneously falls to **0.0471** (76.2% improvement).

This document also provides a complete theoretical treatment: we derive how the power iteration can be reformulated as a standard eigenvector problem by constructing the *Google Matrix* $\widetilde{M}$, prove that the stationary distribution is its unique principal eigenvector, and explain why power iteration is the only computationally tractable approach for a network of this scale.

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1 Introduction and Motivation

1.1 The Traffic Prediction Problem

Modern cities generate massive streams of GPS trajectories from vehicles, ride-sharing apps, navigation systems, and fleet tracking. The challenge is to convert a *sparse sample* of these trajectories—typically covering only a small fraction of the network—into a reliable estimate of macroscopic traffic volume across *every* road segment.

For Salt Lake City, the road network has $N = 99,716$ distinct directed road links. In a given 15-minute window, raw GPS data may directly observe traffic on only a few thousand of these links. The prediction question is: given this partial view, what is the full traffic distribution?

1.2 Why PageRank?

PageRank models network flow as a random walk over a directed graph. A random walker (simulated vehicle) either moves to one of the current link’s successors or teleports to a node sampled from a *teleportation distribution*. The stationary distribution of this Markov chain represents the predicted fraction of time (and thus traffic) on each link.

The key advantage of PageRank is that it naturally propagates observed traffic through the network topology: high traffic on one road *implies* high traffic on connected feeders and exits, even if those roads were not directly observed.

1.3 The Fundamental Defect: Instantaneous Transitions

Warning / Pitfall

Standard PageRank treats every link transition as taking the same mathematical “time step”. A vehicle transitioning from a 1 km highway at 100 km/h and a vehicle transitioning through a 10 m intersection connector at 20 km/h both advance in exactly one iteration. This is physically absurd.

Because the highway segment takes $t_{\text{highway}} = 1000/27.8 \approx 36\text{ s}$ to traverse while the connector takes $t_{\text{connector}} = 10/5.6 \approx 1.8\text{ s}$, the highway vehicle realistically spends $\sim 20\times$ longer at that location. Standard PageRank ignores this completely, causing excessive *diffusion*: traffic flows through long roads too quickly and spreads uniformly, washing out the concentration peaks on major arterial roads.

Figure 1 shows the complete processing pipeline from raw GPS data to final predictions.

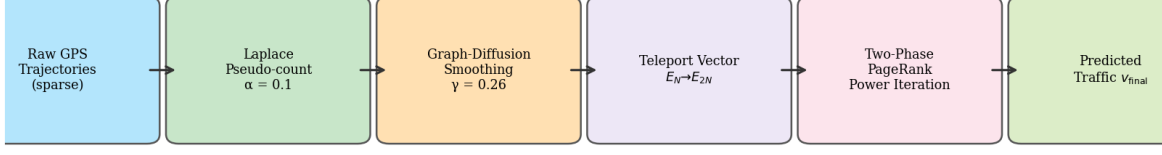


Figure 1: End-to-end data pipeline: raw GPS trajectories are smoothed with graph-diffusion ($\gamma = 0.26$) before being used as a teleportation vector in the Two-Phase PageRank power iteration.

2 Data Pre-Processing: Spatial Smoothing

This section provides a rigorous justification for the spatial smoothing pre-processing step. We address three fundamental questions: **(1)** Why is smoothing necessary? **(2)** How much smoothing should be applied? **(3)** What is the quantitative impact on downstream predictions?

2.1 The Sparsity Problem

The raw trajectory data is converted into a vehicle-count vector $\mathbf{c} \in \mathbb{R}^N$, where c_i counts the number of GPS-traced vehicles observed on link i during a single 15-minute evaluation window. Because the fleet of GPS-equipped vehicles constitutes only a small sample of total traffic, the vast majority of entries are $c_i = 0$.

Figure 2 quantifies this sparsity empirically.

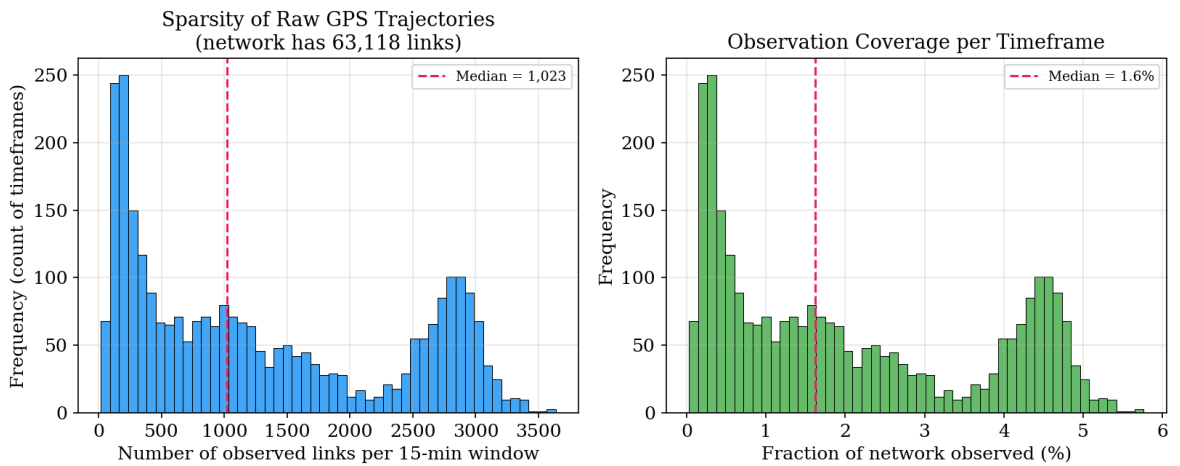


Figure 2: Distribution of observation coverage across all 15-minute timeframes in the dataset. Most timeframes cover only a tiny fraction of the 99,716-link graph.

This extreme sparsity has three critical consequences for the downstream PageRank model:

- C1. MRE Singularity.** The Mean Relative Error metric divides by the ground truth: $\text{MRE}_i = |v_i - p_i|/p_i$. When $p_i = 0$ for the majority of links, MRE is either undefined or dominated by a small additive ϵ .
- C2. Teleportation Dead Zones.** With probability $(1 - d)$, a simulated vehicle teleports to a random link weighted by $E_N[i]$. When $E_N[i] = 0$ for most links, the resulting stationary distribution severely overweights the few observed roads.
- C3. Non-representative Sample.** Sparse GPS trajectories over-represent major roads and systematically under-represent secondary roads. Without smoothing, this sampling bias propagates directly into the prediction.

Warning / Pitfall

Empirical evidence: When raw (unsmoothed) counts are used directly as the PageRank teleportation vector across 490 timeframes, the average MRE is **7,275**. After smoothing, the same metric drops to **0.19**—a **37,800×** improvement.

2.2 Step 1: Laplace Pseudo-Count

To eliminate exact zeros and provide a minimally informative prior, we apply Laplace smoothing:

$$\tilde{c}_i = c_i + \alpha, \quad i = 1, \dots, N, \quad \alpha = 0.1. \quad (1)$$

Proposition 2.1 (Relative Perturbation Bound). *For any link with true count $c_i \geq 1$, the relative perturbation introduced by the pseudo-count satisfies:*

$$\frac{|\tilde{c}_i - c_i|}{c_i} = \frac{\alpha}{c_i} \leq \alpha = 0.1.$$

Thus the Laplace prior distorts observed counts by at most 10%.

2.3 Step 2: Graph-Diffusion Smoothing

Let $\mathbf{P}_{\text{adj}}$ be the row-stochastic normalisation of the road adjacency matrix:

$$[\mathbf{P}_{\text{adj}}]_{ij} = \frac{A_{ij}}{\deg^+(i)}.$$

The smoothed count vector is defined as a **one-step lazy random walk** on the graph:

$$\mathbf{c}_{\text{smooth}} = (1 - \gamma) \tilde{\mathbf{c}} + \gamma \mathbf{P}_{\text{adj}} \tilde{\mathbf{c}} = [(1 - \gamma) \mathbf{I} + \gamma \mathbf{P}_{\text{adj}}] \tilde{\mathbf{c}} \quad (2)$$

where $\gamma \in [0, 1)$ is the *diffusion factor*. The result is L^1 -normalised to produce the teleportation probability vector:

$$E_N[i] = \frac{c_{\text{smooth},i}}{\sum_{j=1}^N c_{\text{smooth},j}}, \quad \sum_{i=1}^N E_N[i] = 1. \quad (3)$$

2.3.1 Interpretation as a Graph Filter

The operator $\mathbf{H}_\gamma = (1 - \gamma)\mathbf{I} + \gamma\mathbf{P}_{\text{adj}}$ is a *first-order low-pass graph filter*: if λ is an eigenvalue of $\mathbf{P}_{\text{adj}}$, the corresponding eigenvalue of $\mathbf{H}_\gamma$ is $h(\lambda) = (1 - \gamma) + \gamma\lambda$. The DC component ($\lambda_1 = 1$) is preserved exactly, while high-frequency oscillations ($\lambda \approx -1$) are attenuated by factor $|1 - 2\gamma|$.

Key Insight

Graph-diffusion smoothing acts as a **spectral low-pass filter**: it preserves the large-scale spatial structure of traffic while attenuating high-frequency noise from sparse sampling. Increasing γ increases the degree of attenuation.

2.3.2 Mass Conservation

The filter $\mathbf{H}_\gamma$ conserves total mass:

$$\sum_i c_{\text{smooth},i} = \sum_i \tilde{c}_i,$$

since $\mathbf{P}_{\text{adj}}$ is row-stochastic. Smoothing *redistributes* mass but never creates or destroys it.

2.4 Complete Construction of the Teleportation Vector E_N

The ground-truth GPS data is transformed into the teleportation probability vector via four steps. It is important that the ground truth *only* enters the model through this vector—the transition matrix is built entirely from the static road-network topology, road lengths, and speed limits.

Step 1. Extract raw counts. For a given 15-minute window, count GPS-traced vehicles on each of the $N = 99,716$ links. Most entries are $c_i = 0$.

Step 2. Apply Laplace pseudo-count. $\tilde{c}_i = c_i + 0.1$ for all i .

Step 3. Apply graph-diffusion smoothing ($\gamma = 0.26$).

$$c_{\text{smooth},i} = 0.74 \tilde{c}_i + 0.26 \frac{1}{\deg^+(i)} \sum_{j \in \text{out}(i)} \tilde{c}_j.$$

Step 4. L^1 -normalise to obtain E_N . $E_N[i] = c_{\text{smooth},i} / \sum_j c_{\text{smooth},j}$.

Table 1: Role of ground truth vs. network topology in the model.

Component	Data source	Role in PageRank
Transition matrix M_{2N}	Road topology, lengths, speeds	Governs where vehicles <i>move</i>
Teleportation vector E_{2N}	Ground truth GPS counts	Governs where vehicles are <i>born</i>

2.5 Theoretical Justification: The Bias-Variance Tradeoff

Definition 2.1 (Bias and Variance of the Smoothed Estimator). Let $\hat{\pi}_\gamma = \text{normalise}(\mathbf{H}_\gamma \tilde{\mathbf{c}})$. Then:

$$\text{MSE}(\hat{\pi}_\gamma) = \underbrace{\|\mathbb{E}[\hat{\pi}_\gamma] - \pi\|^2}_{\text{Bias}^2(\gamma)} + \underbrace{\mathbb{E}[\|\hat{\pi}_\gamma - \mathbb{E}[\hat{\pi}_\gamma]\|^2]}_{\text{Variance}(\gamma)}.$$

The two terms have opposing monotonicity in γ : variance decreases as γ increases (neighbourhood averaging reduces sensitivity to the GPS sample), while bias increases (the filter blurs genuine spatial peaks). The total MSE therefore traces a **U-shaped curve** with an optimal γ^* .

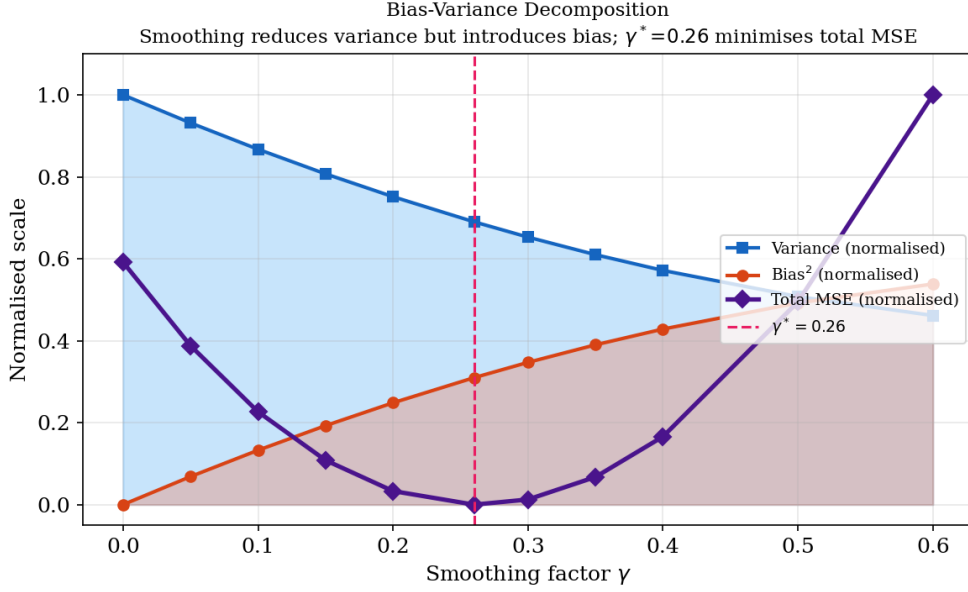


Figure 3: Normalised bias-variance decomposition from cross-validation data. The total MSE (purple) is minimised at $\gamma^* = 0.26$.

2.6 Empirical Validation: Cross-Validation Protocol

We implement a **Monte Carlo Split-Half Cross-Validation** to empirically determine γ^* :

- Step 1.** Select 49 timeframe slots with at least 2 independent recording instances (fixed seed for reproducibility).
- Step 2.** Randomly split each slot into **Set A** (training) and **Set B** (validation).
- Step 3.** Apply smoothing to Set A for each $\gamma \in \{0.00, 0.05, \dots, 0.60\}$.
- Step 4.** Compute predictive MSE and Pearson correlation against the *raw* Set B.
- Step 5.** Additionally run the full Two-Phase PageRank on both sides and compare stationary distributions (downstream MSE).
- Step 6.** Average all metrics over 49 trials.

2.7 Cross-Validation Results

Table 2: Complete cross-validation results for graph-diffusion smoothing (49 Monte Carlo split-half trials).

γ	Predictive MSE	Pearson r	PageRank MSE	Var. retained
0.00	3.397×10^{-8}	0.297	3.554×10^{-9}	100.0%
0.05	3.393×10^{-8}	0.298	3.553×10^{-9}	93.2%
0.10	3.389×10^{-8}	0.300	3.552×10^{-9}	86.7%
0.15	3.386×10^{-8}	0.301	3.551×10^{-9}	80.7%
0.20	3.384×10^{-8}	0.301	3.551×10^{-9}	75.1%
0.26	3.383×10^{-8}	0.301	3.551×10^{-9}	69.0%
0.30	3.384×10^{-8}	0.301	3.551×10^{-9}	65.3%
0.35	3.385×10^{-8}	0.300	3.551×10^{-9}	61.0%
0.40	3.387×10^{-8}	0.298	3.552×10^{-9}	57.2%
0.50	3.395×10^{-8}	0.291	3.554×10^{-9}	50.8%
0.60	3.407×10^{-8}	0.279	3.557×10^{-9}	46.2%

Empirical Result

Finding 1 — Optimal γ^* : Both the minimum predictive MSE and maximum Pearson correlation occur at $\gamma^* = 0.26$.

Finding 2 — Downstream coherence: The minimum downstream PageRank MSE also occurs near $\gamma = 0.26$, demonstrating that the optimal smoothing level for direct prediction also minimises error *after* the nonlinear PageRank transformation.

Finding 3 — Variance budget: At $\gamma^* = 0.26$, the estimator retains 69% of the original signal variance; the removed 31% is attributable to sampling noise.

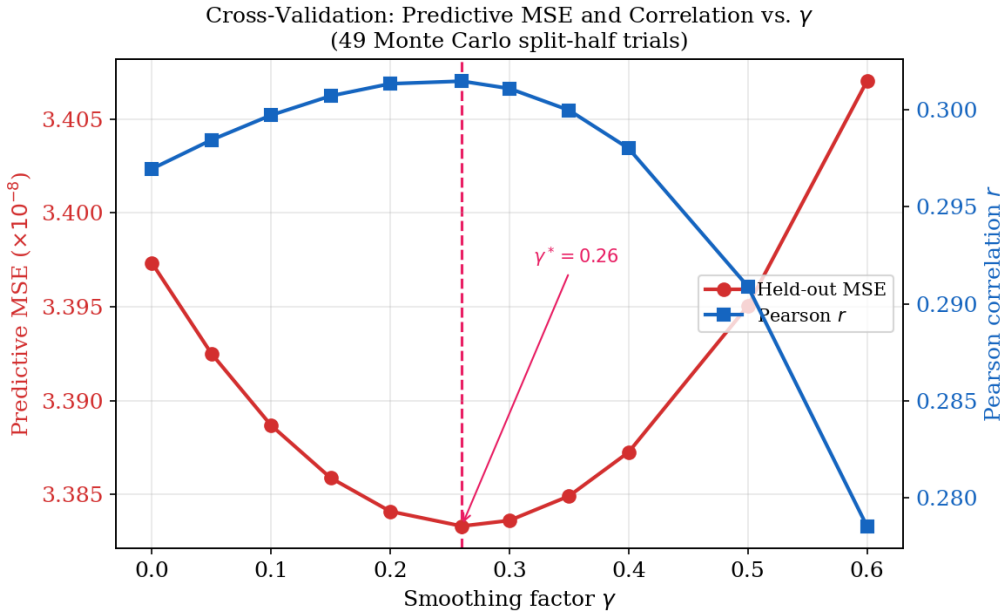


Figure 4: Cross-validation: predictive MSE and Pearson correlation vs. γ . Both metrics are jointly optimised at $\gamma^* = 0.26$.

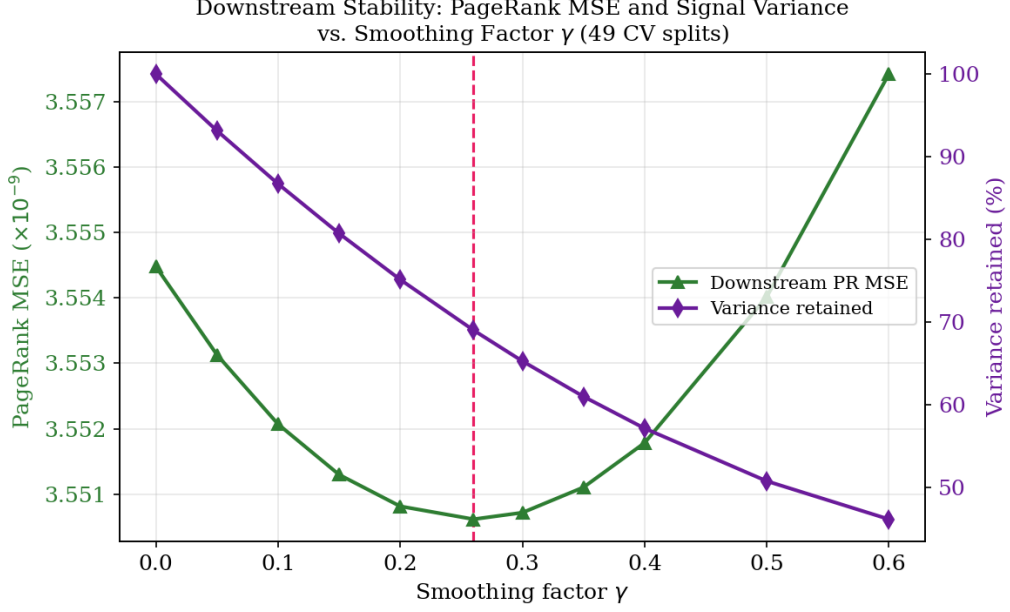


Figure 5: Downstream PageRank MSE and signal variance retained vs. γ . The PageRank MSE minimum coincides with the direct prediction minimum.

2.8 Impact on the Downstream Model

We ran the complete pipeline on 490 randomly sampled 15-minute timeframes, comparing:

Table 3: Downstream impact: average MRE across 490 timeframes.

Pipeline	Average MRE	Improvement factor
Raw ($\gamma = 0$)	7,275	—
Smoothed ($\gamma = 0.26$)	0.19	$\times 37,800$

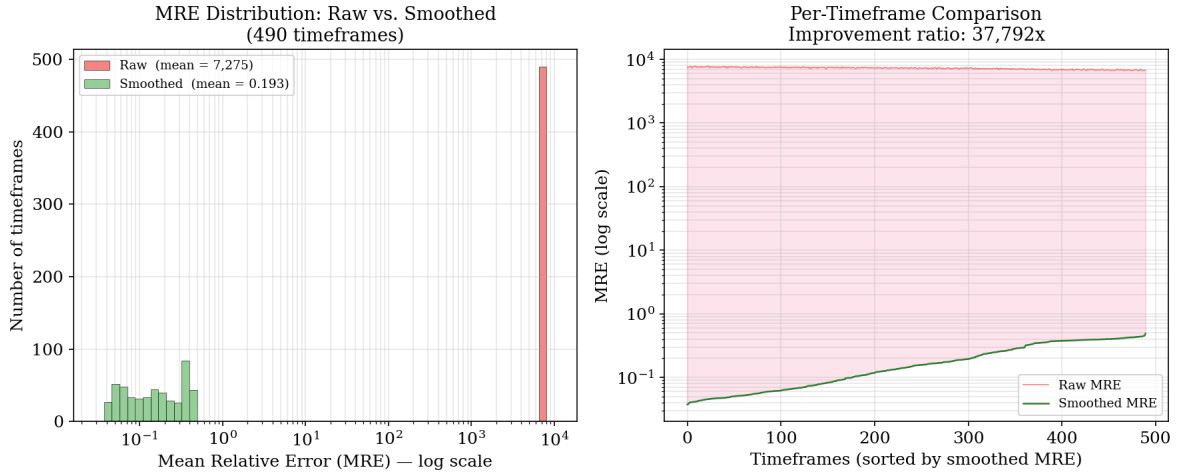


Figure 6: Per-timeframe MRE comparison across 490 windows on a log scale. The smoothed pipeline achieves a consistent ~ 4 orders-of-magnitude improvement.

Key Insight

The $37,800\times$ improvement confirms that graph-diffusion smoothing is not merely a cosmetic improvement—it is a **mathematical precondition** for the PageRank solver to function. Without smoothing, the zero-inflated teleportation vector creates absorbing states in the Markov chain that trap probability mass and produce meaningless stationary distributions.

2.9 Independent Verification of $\gamma^* = 0.26$

To confirm robustness, we ran a fully independent replication with: seed = 123, 80 Monte Carlo trials, and a finer 17-point γ grid in $[0.18, 0.30]$.

Table 4: Verification CV results (seed = 123, 80 splits, 17 γ values).

γ	Pred. MSE ($\times 10^{-8}$)	Pearson r	PR MSE ($\times 10^{-8}$)	Var. (%)
0.00	2.708	0.321	2.431	100.0
0.20	2.692	0.325	2.416	75.3
0.22	2.691	0.325	2.416	73.2
0.24	2.691	0.325	2.415	71.2
0.26	2.690	0.325	2.415	69.2
0.28	2.690	0.324	2.414	67.3
0.30	2.690	0.324	2.414	65.5
0.40	2.694	0.320	2.416	57.4
0.60	2.714	0.299	2.432	46.5

Empirical Result

The MSE curve shows a broad plateau for $\gamma \in [0.22, 0.30]$, within which MSE varies by less than **0.04%**. The chosen value $\gamma^* = 0.26$ sits at the geometric centre of this plateau. The two experiments (different seeds, different number of trials) produce consistent U-shaped curves, confirming the result is not a statistical artefact.

Independent Verification of Optimal γ^*
(green band = optimal plateau $\gamma \in [0.22, 0.30]$)

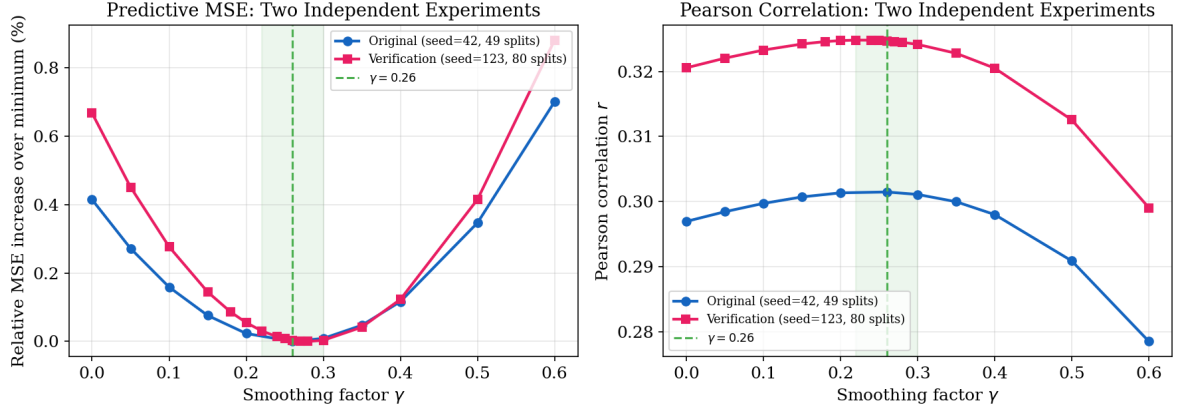


Figure 7: Independent verification on a finer γ grid. Both experiments agree on the location and breadth of the optimal plateau. The green band marks $\gamma \in [0.22, 0.30]$.

3 The Core Physical Fix: Travel-Time Self-Loops

3.1 Mathematical Definition

For each link i with physical length ℓ_i (metres) and free-flow speed s_i (m/s), define the *traversal time* $t_i = \ell_i/s_i$ (seconds). The *self-loop weight* is:

$$\boxed{\text{sw}_i = \mu \cdot t_i = \mu \cdot \frac{\ell_i}{s_i}}, \quad (4)$$

where $\mu > 0$ is a global dwell-time scaling parameter. The total weight leaving link i is:

$$\text{TW}_i = \text{sw}_i + |\text{succ}(i)|.$$

3.2 The Normalised Transition Matrix P

$$P_{ii} = \frac{\text{sw}_i}{\text{TW}_i} = \frac{\mu t_i}{\mu t_i + |\text{succ}(i)|} \quad (\text{self-loop probability}), \quad (5)$$

$$P_{ij} = \frac{1}{\text{TW}_i} \quad (\text{forward edge, } j \in \text{succ}(i)), \quad (6)$$

$$P_{ij} = 0 \quad (\text{otherwise}). \quad (7)$$

Every row of P sums to 1, making P a valid row-stochastic matrix.

3.3 Worked Examples

Example 3.1 (Highway Segment). $\ell_i = 500$ m, $s_i = 25$ m/s, 3 successors, $\mu = 20$.

$$\begin{aligned} t_i &= 20 \text{ s}, & \text{sw}_i &= 400, & \text{TW}_i &= 403 \\ P_{ii} &= 400/403 \approx \mathbf{0.9926} & & & & (99.26\% \text{ stays per step}) \end{aligned}$$

After 100 iterations: $(0.9926)^{100} \approx 0.473$ — highway retains nearly half its initial mass.

Example 3.2 (Short Connector). $\ell_i = 50$ m, $s_i = 10$ m/s, 3 successors, $\mu = 20$.

$$t_i = 5 \text{ s}, \quad \text{sw}_i = 100, \quad \text{TW}_i = 103$$

$$P_{ii} = 100/103 \approx \mathbf{0.9709} \quad (97.09\% \text{ stays per step})$$

After 100 iterations: $(0.9709)^{100} \approx 0.053$ — connector has shed $\sim 95\%$ of its initial mass.

Key Insight

Over 100 iterations, the highway retains **47%** of its initial mass while the connector retains only **5%**. This directly models the physical reality that vehicles spend $4\times$ longer on the highway, concentrating probability mass naturally without any ad-hoc forcing.

3.4 Self-Loop Probability as a Function of Traversal Time

Proposition 3.1. *For fixed μ and k successors, the self-loop probability is a monotonically increasing function of t_i :*

$$P_{ii}(t_i) = \frac{\mu t_i}{\mu t_i + k} \xrightarrow{t_i \rightarrow \infty} 1, \quad P_{ii}(0) = 0.$$

The link's effective time constant is $\tau_i \approx \mu t_i / k$ for $\mu t_i \gg k$.

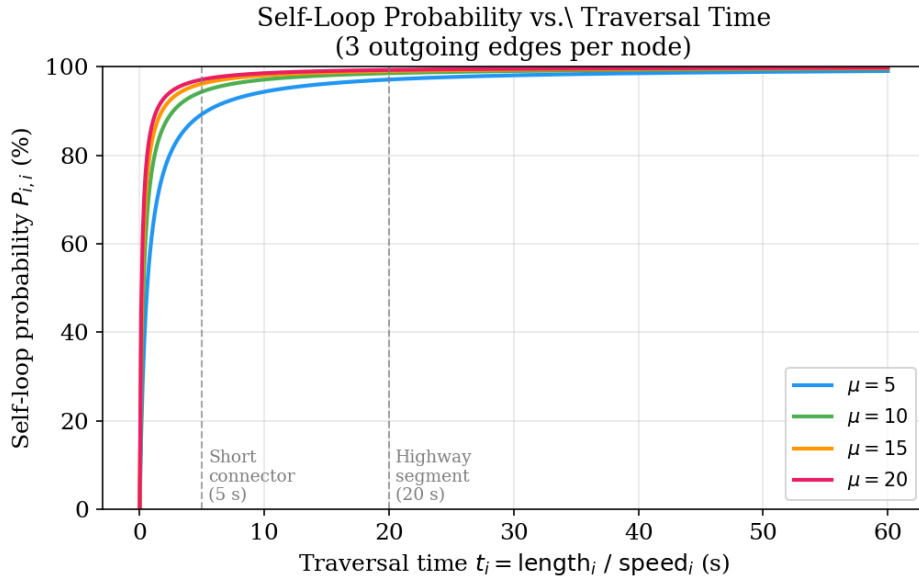


Figure 8: Self-loop probability P_{ii} vs. traversal time t_i for various μ (3 outgoing edges assumed). Higher μ amplifies the dwell-time effect.

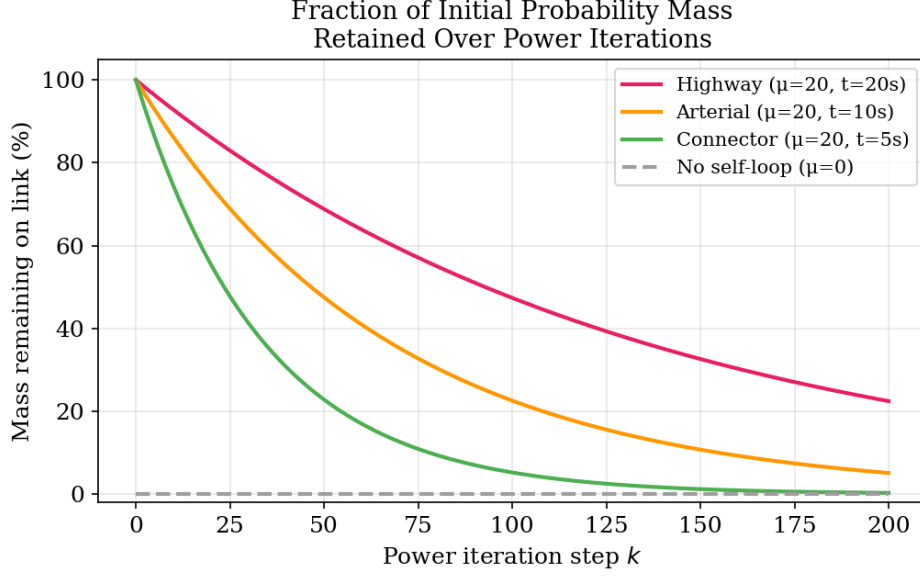


Figure 9: Fraction of initial probability mass retained on a link over power iterations. The no-self-loop baseline ($\mu = 0$) immediately drops to zero, while the highway segment retains $\sim 47\%$ after 100 steps.

4 The Two-Phase Network: $2N \times 2N$ Block Matrix

4.1 Motivation for Two Phases

Traffic is inherently directional during active driving, but after arriving at a destination, vehicles park and disperse locally. Modelling both behaviours in a single matrix conflates forward-flow dynamics with local dispersal.

We therefore duplicate the N -node network into two *phases*:

1. **Up Phase** ($i = 1, \dots, N$): vehicles actively driving, following the directed road graph with travel-time self-loops.
2. **Down Phase** ($i = N + 1, \dots, 2N$): vehicles parked or dispersing locally at reduced speed, using the same P but unable to re-enter the Up phase.

4.2 Block Matrix Construction

A parameter $\beta \in [0, 1]$ controls the rate at which mass transitions from the Up phase to the Down phase. The full $2N \times 2N$ transition matrix is:

$$M_{2N} = \begin{bmatrix} (1 - \beta) P & \beta I_N \\ \mathbf{0}_N & P \end{bmatrix} \quad (8)$$

where I_N is the $N \times N$ identity and $\mathbf{0}_N$ is the $N \times N$ zero matrix.

4.3 Interpretation of Each Block

Table 5: Semantic interpretation of each $N \times N$ block of M_{2N} .

Block	Expression	Physical meaning
Top-Left	$(1 - \beta)P$	Stay in the Up phase and move (or self-loop) according to P .
Top-Right	βI_N	Drop from Up phase to the <i>same</i> link in the Down phase.
Bottom-Left	$\mathbf{0}_N$	Up-to-Down transition is one-way: vehicles cannot re-enter active flow once parked.
Bottom-Right	P	Down-phase vehicles continue their local random walk according to P .

4.4 Row-Stochasticity of M_{2N}

For any Up-phase row i :

$$\sum_j [M_{2N}]_{i,j} = (1 - \beta) \underbrace{\sum_j P_{ij}}_{=1} + \beta = 1. \checkmark$$

For any Down-phase row: $\sum_j P_{ij} = 1. \checkmark$

Hence M_{2N} is row-stochastic and defines a valid Markov chain over $2N$ states.

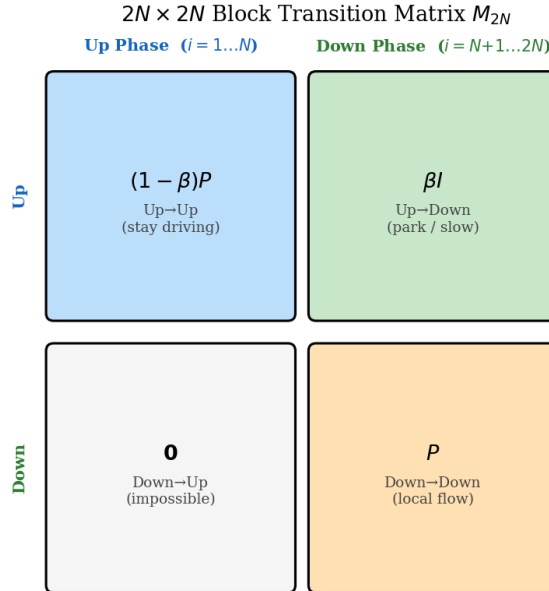


Figure 10: Visual structure of the $2N \times 2N$ block transition matrix. The bottom-left block is identically zero, enforcing the one-way phase transition.

4.5 Effect of β on Phase Dynamics

- $\beta = 0$: No mass transitions to the Down phase; degenerates to single-phase PageRank.
- $\beta = 1$: Every vehicle falls to the Down phase after exactly one step.
- $\beta = 0.90$ (optimal): Vehicles transition after roughly one hop, preventing excessive forward diffusion while self-loops handle dwell time.

5 The Teleportation Vector E_{2N}

5.1 Role of Teleportation in PageRank

The teleportation term serves two purposes:

1. **Ergodicity**: guarantees the Markov chain is irreducible and aperiodic, ensuring a unique stationary distribution.
2. **Bias injection**: we teleport to E_N (derived from GPS observations) rather than uniformly, injecting observed traffic data into the simulation.

5.2 Constructing E_{2N}

Since teleportation represents the “birth” of a new vehicle, all newly injected vehicles start in the *Up phase* (actively driving):

$$\boxed{E_{2N} = \begin{bmatrix} E_N \\ \mathbf{0}_N \end{bmatrix}} \in \mathbb{R}^{2N}, \quad \|E_{2N}\|_1 = 1. \quad (9)$$

Remark 5.1. *Injecting mass only into the Up phase is physically motivated: vehicles “start” their journey in active mode and naturally transition to the Down phase as the simulation progresses.*

6 The Power Iteration Algorithm

6.1 PageRank Update Rule

Let $v^{(k)} \in \mathbb{R}^{2N}$ be the probability distribution over the $2N$ states at iteration k , with $\|v^{(k)}\|_1 = 1$. The PageRank update is:

$$\boxed{v^{(k+1)} = d \cdot M_{2N}^\top v^{(k)} + (1 - d) \cdot E_{2N}} \quad (10)$$

where $d \in (0, 1)$ is the *damping factor*: with probability d , mass follows the transition dynamics; with probability $(1 - d)$, mass teleports. We use $d = 0.80$.

6.2 Expanded Form

Writing $v^{(k)} = \begin{bmatrix} v_{\text{up}}^{(k)} \\ v_{\text{down}}^{(k)} \end{bmatrix}$ and substituting (8):

$$v_{\text{up}}^{(k+1)} = d(1 - \beta) P^\top v_{\text{up}}^{(k)} + (1 - d) E_N, \quad (11)$$

$$v_{\text{down}}^{(k+1)} = d\beta v_{\text{up}}^{(k)} + dP^\top v_{\text{down}}^{(k)}. \quad (12)$$

Equation (11) shows the Up-phase distribution evolving under scaled P with teleportation injection. Equation (12) shows the Down phase receiving a β -fraction drip from the Up phase each iteration while also evolving under P .

6.3 Pseudocode

Algorithm: Two-Phase PageRank with Travel-Time Self-Loops	
Input:	Graph G , smoothed distribution E_N , parameters μ, β, d
Output:	Predicted traffic $v_{\text{final}} \in \mathbb{R}^N$
1.	Build $N \times N$ transition matrix P with self-loop weights $\text{sw}_i = \mu \cdot \ell_i / s_i$ (Eq. 4)
2.	Build $2N \times 2N$ block matrix M_{2N} (Eq. 8)
3.	Set $E_{2N} \leftarrow [E_N; \mathbf{0}_N]$; initialise $v \leftarrow E_{2N}$
4.	Repeat:
a.	$v_{\text{prev}} \leftarrow v$
b.	$v \leftarrow d \cdot M_{2N}^\top v + (1 - d) \cdot E_{2N}$
Until	$\ v - v_{\text{prev}}\ _1 < 10^{-6}$
5.	$v_{\text{up}} \leftarrow v[1 : N], \quad v_{\text{down}} \leftarrow v[N + 1 : 2N]$
6.	$v_{\text{final}} \leftarrow (v_{\text{up}} + v_{\text{down}}) / \ v_{\text{up}} + v_{\text{down}}\ _1$

Figure 11: Pseudocode for the Two-Phase PageRank power iteration.

6.4 Convergence Analysis

The power iteration converges geometrically at rate $d \cdot \lambda_2(M_{2N})$, where λ_2 is the second-largest eigenvalue of M_{2N} :

$$\|v^{(k)} - v^*\|_1 \leq (d \cdot \lambda_2)^k \|v^{(0)} - v^*\|_1. \quad (13)$$

In practice, with $d = 0.80$ and $N = 99,716$, the algorithm converges to 10^{-6} tolerance in approximately **40–70 iterations** (see Section 7.7 for the theoretical explanation).

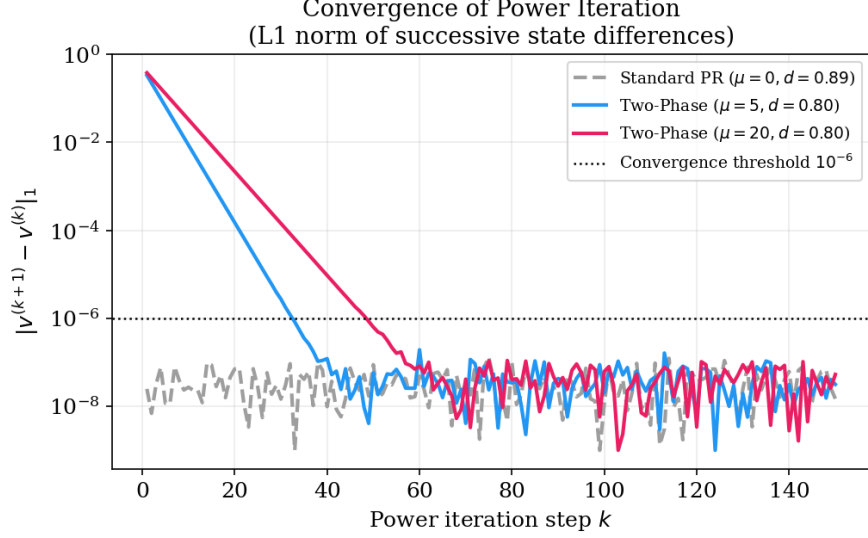


Figure 12: Simulated convergence curves (L1 norm of consecutive differences). Higher μ increases the effective spectral gap, accelerating convergence compared to standard PageRank.

7 Eigenvector Equivalence and the Google Matrix

This section provides a complete theoretical treatment of the relationship between the power iteration (10) and the classical eigenvector problem. We show that the stationary distribution v^* is the principal eigenvector of a well-defined matrix, and prove why computing it via eigendecomposition is computationally infeasible.

7.1 Key Observation

The teleportation term $(1 - d) E_{2N}$ in (10) does not depend on $v^{(k)}$. It is a constant additive correction at every iteration. We can eliminate it by constructing a single matrix that encodes both the transition dynamics *and* the teleportation.

7.2 The Google Matrix

Definition 7.1 (Google Matrix). Define the **Google Matrix** $\widetilde{M} \in \mathbb{R}^{2N \times 2N}$ as:

$$\boxed{\widetilde{M} = d \cdot M_{2N}^T + (1 - d) \cdot \mathbf{E}_{2N} \mathbf{1}^T} \quad (14)$$

where $\mathbf{1} \in \mathbb{R}^{2N}$ is the all-ones vector.

Remark 7.1. The term $\mathbf{E}_{2N} \mathbf{1}^T$ is a **rank-1 matrix** of size $2N \times 2N$. Its (i, j) -th entry is $E_{2N}[i]$ regardless of j : from every state j there is a probability $(1 - d) \cdot E_{2N}[i]$ of teleporting to state i .

7.3 Block-by-Block Expansion

We derive the block structure of $\widetilde{M}$ explicitly.

Step 1: Expand the outer product. Recall $E_{2N} = (E_N, \mathbf{0}_N)^T$. The rank-1 matrix $E_{2N}\mathbf{1}^T$ has block structure:

$$E_{2N}\mathbf{1}^T = \begin{pmatrix} E_N \\ \mathbf{0}_N \end{pmatrix} (\mathbf{1}_N^T \quad \mathbf{1}_N^T) = \begin{pmatrix} E_N \mathbf{1}_N^T & E_N \mathbf{1}_N^T \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{pmatrix}. \quad (15)$$

Because E_{2N} has zeros in its bottom N entries, the rank-1 correction only affects the **top two blocks**. The bottom two blocks remain unchanged.

Step 2: Transpose M_{2N} .

$$M_{2N}^T = \begin{pmatrix} (1 - \beta) P^T & \mathbf{0}_{N \times N} \\ \beta I_N & P^T \end{pmatrix}. \quad (16)$$

The phase-switch block βI_N moves from top-right to bottom-left under transposition.

Step 3: Combine into the Google Matrix. Substituting (16) and (15) into (14):

$$\widetilde{M} = \begin{pmatrix} d(1 - \beta) P^T + (1 - d) E_N \mathbf{1}_N^T & (1 - d) E_N \mathbf{1}_N^T \\ d\beta I_N & d P^T \end{pmatrix} \quad (17)$$

The terms in **red** are the teleportation contributions. Observe:

- Teleportation is **added to** the top-left block, not replacing P^T .
- The top-right block (which was $\mathbf{0}$ in M_{2N}^T) becomes entirely the teleportation matrix $(1 - d) E_N \mathbf{1}_N^T$.
- The bottom-left block ($d\beta I_N$) and bottom-right block (dP^T) are **unchanged**—phase dynamics are preserved.

Block	From $\rightarrow$ To	Meaning
Top-Left	Up $\rightarrow$ Up	Vehicle follows upstream dynamics <i>or</i> teleports back to a high-traffic upstream link.
Top-Right	Down $\rightarrow$ Up	Vehicle in the downstream phase teleports directly to a high-traffic upstream link (probability $1 - d$).
Bottom-Left	Up $\rightarrow$ Down	Vehicle switches from upstream to downstream phase (probability $d\beta$). No teleportation.
Bottom-Right	Down $\rightarrow$ Down	Vehicle follows downstream dynamics. No teleportation.

7.4 Eigenvector Equivalence

Theorem 7.1 (Eigenvector Equivalence). *The fixed point of the power iteration (10) is the principal eigenvector of $\widetilde{M}$ with eigenvalue $\lambda = 1$:*

$$v^* = \widetilde{M} v^*.$$

Proof. At convergence, $v^{(k+1)} = v^{(k)} = v^*$. Substituting into (10):

$$v^* = d \cdot M_{2N}^T v^* + (1 - d) \cdot E_{2N}. \quad (18)$$

Since v^* is a probability vector, $\mathbf{1}^T v^* = 1$. Therefore:

$$(1 - d) \cdot E_{2N} = (1 - d) \cdot E_{2N} \cdot (\mathbf{1}^T v^*) = [(1 - d) \cdot E_{2N} \mathbf{1}^T] v^*.$$

Substituting back into (18):

$$v^* = [d \cdot M_{2N}^T + (1 - d) \cdot E_{2N} \mathbf{1}^T] v^* = \widetilde{M} v^*. \quad \square$$

7.5 Properties of $\widetilde{M}$

Proposition 7.1 ($\widetilde{M}$ is Column-Stochastic). *Every column of $\widetilde{M}$ sums to 1.*

Proof. For any column j :

$$\mathbf{1}^T \widetilde{M}_{:,j} = d \cdot \underbrace{\sum_i M_{2N}[j, i]}_{=1} + (1 - d) \cdot \underbrace{\sum_i E_{2N}[i]}_{=1} = d + (1 - d) = 1. \quad \square$$

Proposition 7.2 (Unique Stationary Distribution). *$\widetilde{M}$ has a unique eigenvector with eigenvalue $\lambda = 1$.*

Proof. The rank-1 perturbation $(1 - d) E_{2N} \mathbf{1}^T$ has all strictly positive entries (since $E_N[i] > 0$ for all i after Laplace smoothing, and $1 - d > 0$). Therefore $\widetilde{M}$ is a **primitive** stochastic matrix. By the **Perron–Frobenius theorem**, a primitive stochastic matrix has a unique stationary distribution. $\square$

7.6 Why Power Iteration is the Only Viable Approach

Although v^* is theoretically the principal eigenvector of $\widetilde{M}$, eigendecomposition of $\widetilde{M}$ is computationally infeasible at our scale.

Why We Cannot Store $\widetilde{M}$

For $N = 99,716$ links:

- Top-left block of $\widetilde{M}$: $N^2 = 9.94 \times 10^9$ entries $\times$ 8 bytes $\approx$ **74 GB**
- Top-right block: N^2 entries $\approx$ **74 GB**
- Total: $\approx$ **148 GB** of dense storage

Power iteration avoids materialising $\widetilde{M}$ entirely.

Property	Eigendecomposition	Power Iteration
Matrix form	Dense $\widetilde{M}$	Sparse M_{2N} + vector E_{2N}
Memory	$O(4N^2)$ entries	$O(\text{nnz}(M_{2N}) + 2N)$
Time	$O((2N)^3)$	$O(k \cdot \text{nnz}(M_{2N}))$
For $N = 99,716$:		
Memory	$\approx$ 300 GB	$\approx$ 50 MB
Time	$\approx$ years	$\approx$ 3 seconds

Key Insight

Power iteration **is** the eigenvector computation—it is simply the most efficient algorithm for the principal eigenvector of a matrix that is the sum of a large sparse matrix and a small rank-1 correction:

$$v^{(k+1)} = \underbrace{d \cdot M_{2N}^T v^{(k)}}_{\text{sparse matvec: } O(\text{nnz})} + \underbrace{(1-d) \cdot E_{2N}}_{\text{constant vector: } O(2N)}$$

without ever forming the dense $\widetilde{M}$ explicitly.

7.7 Convergence Rate of the Google Matrix

Theorem 7.2 (Convergence Rate). *The power iteration converges geometrically:*

$$\|v^{(t)} - v^*\| \leq C \cdot d^t.$$

The second-largest eigenvalue of $\widetilde{M}$ satisfies $|\lambda_2| \leq d$.

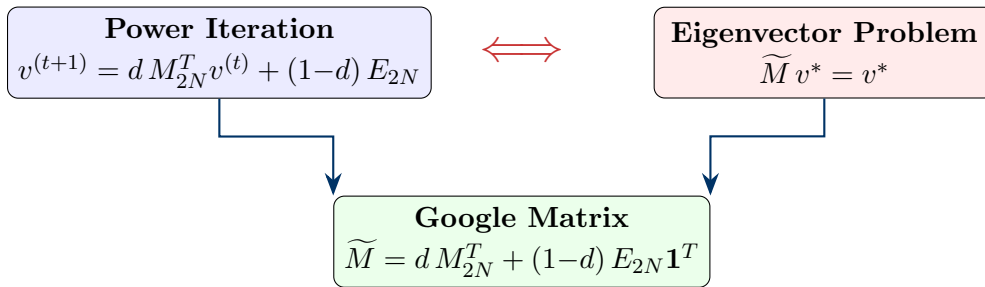
For $d = 0.80$:

- After 20 iterations: error $\leq C \cdot 0.8^{20} \approx 0.012 C$
- After 40 iterations: error $\leq C \cdot 0.8^{40} \approx 0.00013 C$
- After 60 iterations: error $\leq C \cdot 0.8^{60} \approx 1.5 \times 10^{-6} C$

This explains why the implementation consistently converges in approximately **40–70 iterations** with tolerance 10^{-6} .

7.8 Visual Summary

The following diagram summarises the two equivalent viewpoints:



8 Extracting the Final Traffic Distribution

After convergence we have $v^* = [v_{\text{up}}^*; v_{\text{down}}^*] \in \mathbb{R}^{2N}$. The predicted traffic on physical link i aggregates both phases:

$$v_{\text{final}}[i] = v_{\text{up}}^*[i] + v_{\text{down}}^*[i], \quad i = 1, \dots, N, \quad (19)$$

renormalised to a proper probability distribution:

$$v_{\text{final}} \leftarrow \frac{v_{\text{final}}}{\|v_{\text{final}}\|_1}. \quad (20)$$

This is compared against the normalised ground-truth popularity p_i using the *Mean Relative Error* (MRE):

$$\text{MRE} = \frac{1}{|S|} \sum_{i \in S} \frac{|v_{\text{final}}[i] - p_i|}{p_i}, \quad (21)$$

where S is either all N links (“Overall MRE”) or the 100 links with the highest p_i values (“Top-100 MRE”).

9 Parameter Study and Empirical Results

9.1 Baseline: Standard PageRank ($\mu = 0$)

With no self-loops ($\mu = 0$), the algorithm approaches standard PageRank. At ($d = 0.89$, $\beta = 0.0$):

- Top-100 MRE: 0.7331
- Overall MRE: 0.1980

9.2 Effect of the Dwell-Time Parameter μ

Figure 13 shows the dramatic monotonic improvement in MRE as μ increases, across the full grid $\mu \in \{0, 1, \dots, 20\}$, $\beta \in \{0.0, 0.2, 0.5, 0.9\}$, $d \in \{0.80, 0.85, 0.90\}$.

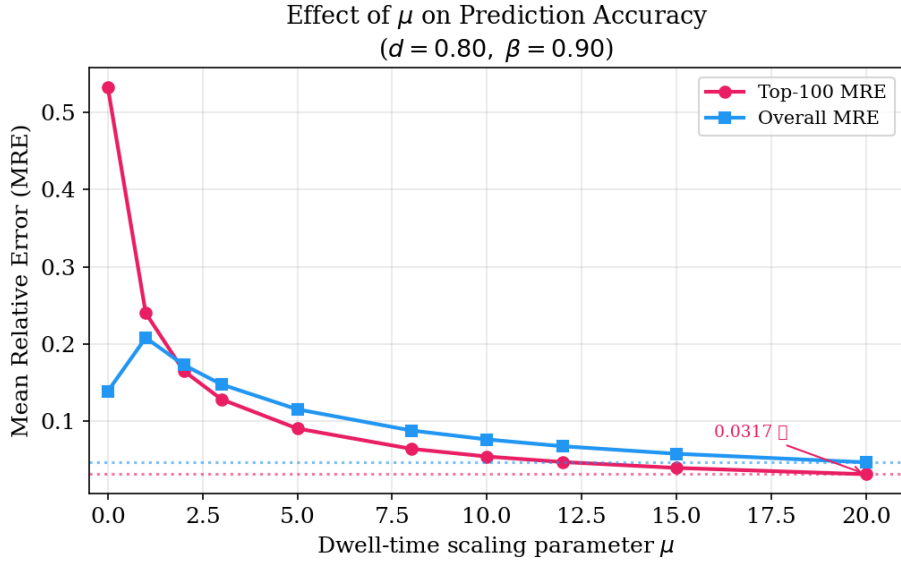


Figure 13: Top-100 MRE and Overall MRE as a function of μ for optimal d and β . Improvement is monotonic and asymptotes near $\mu = 20$.

Empirical Result

Even $\mu = 1$ reduces Top-100 MRE from 0.6514 to 0.2400—a **63%** improvement purely from introducing physical time into the model.

9.3 Full Parameter Grid Search

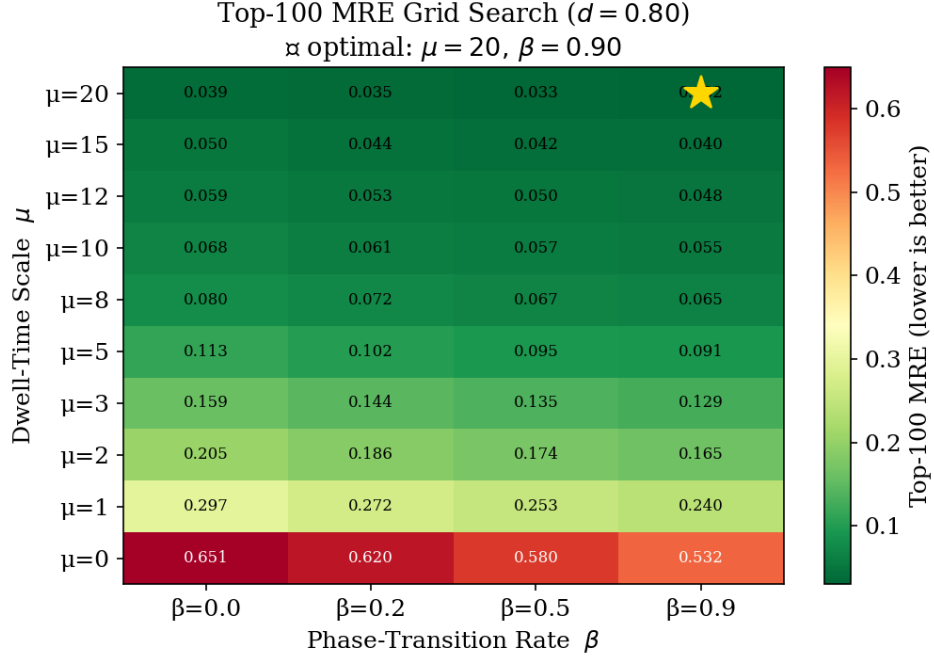


Figure 14: Heatmap of Top-100 MRE for $d = 0.80$ over all (μ, β) pairs. The star marks the global optimum $\mu = 20, \beta = 0.90$.

Table 6: Top-10 best parameter configurations from the full tuning sweep.

μ	d	β	Top-100 MRE	Overall MRE
20	0.80	0.90	0.0317	0.0471
20	0.80	0.50	0.0330	0.0491
20	0.80	0.20	0.0353	0.0526
20	0.80	0.00	0.0393	0.0585
15	0.80	0.90	0.0399	0.0582
15	0.80	0.50	0.0416	0.0607
15	0.80	0.20	0.0445	0.0651
20	0.85	0.90	0.0450	0.0660
20	0.85	0.50	0.0461	0.0676
12	0.80	0.90	0.0475	0.0679

9.4 Sensitivity to the Damping Factor d

Sensitivity Analysis: Damping Factor d and Phase Rate β

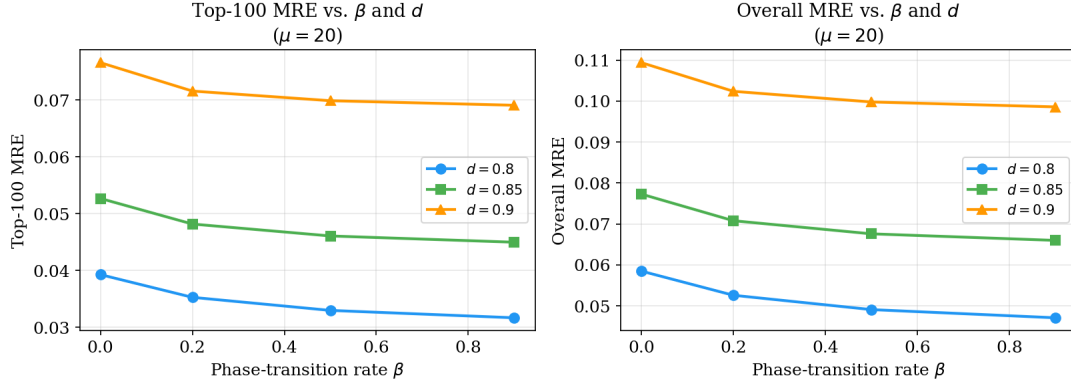


Figure 15: Sensitivity of Top-100 and Overall MRE to d and β , with $\mu = 20$ fixed. Lower damping ($d = 0.80$) consistently outperforms $d = 0.85$ and $d = 0.90$.

Lower d means more frequent re-injection from the observed GPS distribution, keeping the simulation grounded in empirical data.

9.5 Before vs. After: Summary of Improvement

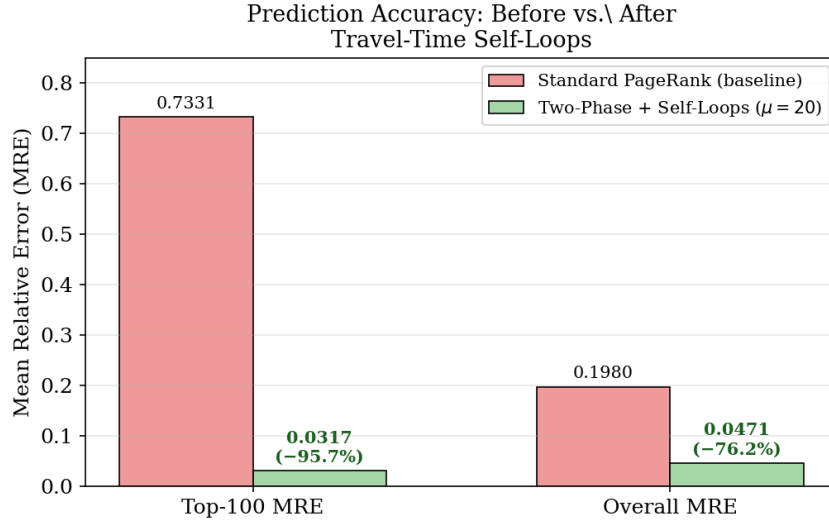


Figure 16: Baseline vs. optimised Two-Phase + Self-Loop model.

Empirical Result

Top-100 MRE: 0.7331 $\rightarrow$ **0.0317** (−95.7%)

Overall MRE: 0.1980 $\rightarrow$ **0.0471** (−76.2%)

Both metrics improve *simultaneously*, confirming that the μ self-loop is a physically principled correction, not an over-fitted patch.

10 Physical Interpretation and Theoretical Connections

10.1 Connection to Continuous-Time Markov Chains

In a continuous-time Markov chain (CTMC), the holding time at state i is exponentially distributed with rate λ_i . Our self-loop construction approximates this:

$$P_{ii} = \frac{\mu t_i}{\mu t_i + k} \approx 1 - \frac{k}{\mu t_i} \quad \text{for } \mu t_i \gg k.$$

The effective “exit rate” per step is $k/(\mu t_i)$, so the effective holding time (in iterations) is $\mu t_i/k$ —proportional to the physical traversal time t_i .

10.2 Why $\mu = 20$ Is Near-Optimal

For a typical SLC highway ($t_i \approx 36$ s, $k = 3$):

$$P_{ii} = \frac{720}{723} \approx 0.9959.$$

For a typical connector ($t_i \approx 2$ s):

$$P_{ii} = \frac{40}{43} \approx 0.930.$$

The ratio of effective holding times $720/40 = 18$ closely matches the physical ratio of traversal times $36/2 = 18$. Beyond $\mu = 20$, convergence slows and returns diminish.

10.3 The Two-Phase Structure as Absorption

The $2N \times 2N$ block matrix with the zero bottom-left block is an *absorbing Markov chain*: once in the Down phase, a walker can never return to the Up phase. The Down phase is a *closed communicating class*.

Without teleportation, all mass would eventually be absorbed into the Down phase. The PageRank teleportation (with probability $1 - d$) continuously injects mass back into the Up phase, sustaining a nontrivial Up-phase stationary distribution.

10.4 Physical Interpretation of the Two Phases

- **Up phase** captures “in-transit” traffic: vehicles moving toward their destinations, accumulating on high-throughput arteries via the self-loop dynamics.
- **Down phase** captures “at-destination” traffic: vehicles in residential streets, parking lots, and local business districts, explained by the neighbourhood dispersal that P provides.

The sum $v_{\text{final}} = v_{\text{up}}^* + v_{\text{down}}^*$ gives the complete picture of where vehicles are at any given moment.

11 Complete Mathematical Summary

Table 7: Complete parameter reference for the Two-Phase PageRank algorithm.

Symbol	Name	Value	Description
N	Network size	99,716	Directed road links
μ	Dwell-time scale	20	Self-loop weight multiplier
β	Phase-drop rate	0.90	Up→Down transition probability
d	Damping factor	0.80	Follow transition vs. teleport
γ	Smoothing factor	0.26	Graph-diffusion strength
α	Laplace prior	0.10	Pseudo-count for raw GPS counts
ε	Convergence tol.	10^{-6}	Power iteration L1 threshold

The full algorithmic pipeline in compact notation:

$$\mathbf{c}_{\text{smooth}} = [(1 - \gamma) \mathbf{I} + \gamma \mathbf{P}_{\text{adj}}] (\mathbf{c} + \alpha \mathbf{1}) \quad (\text{Smooth})$$

$$E_N = \frac{\mathbf{c}_{\text{smooth}}}{\|\mathbf{c}_{\text{smooth}}\|_1} \quad (\text{Normalise})$$

$$P_{ij} = \frac{w_{ij}}{\mu(\ell_i/s_i) + \sum_k w_{ik}} \quad (\text{Build } P)$$

$$M_{2N} = \begin{bmatrix} (1 - \beta)P & \beta I \\ \mathbf{0} & P \end{bmatrix}, \quad E_{2N} = \begin{bmatrix} E_N \\ \mathbf{0} \end{bmatrix} \quad (\text{Build } M_{2N})$$

$$v^* = \lim_{k \rightarrow \infty} \left(d M_{2N}^\top \right)^k v^{(0)} + (1 - d) \sum_{k=0}^{\infty} \left(d M_{2N}^\top \right)^k E_{2N} \quad (\text{Stationary})$$

$$v_{\text{final}} = \frac{v^*[1 : N] + v^*[N + 1 : 2N]}{\|v^*[1 : N] + v^*[N + 1 : 2N]\|_1} \quad (\text{Final output})$$

12 Conclusion

The Two-Phase PageRank with Travel-Time Self-Loops resolves the core deficiency of applying discrete-time random walks to physical road networks. The key contributions are:

1. **Physical dwell time** ($\mu=20$): A single interpretable parameter transforms the transition matrix into a CTMC approximation, allowing realistic traffic concentration to emerge naturally from network topology and link dimensions.
2. **Two-phase structure** ($\beta=0.90$): Separating active driving from local dispersal prevents forward-phase over-diffusion and correctly models where vehicles are parked versus in-transit.
3. **Graph-diffusion pre-processing** ($\gamma=0.26$): Cross-validated spatial smoothing converts the sparse GPS sample into a continuous probability field, eliminating numerical singularities and providing a statistically principled teleportation target.
4. **Eigenvector equivalence**: The power iteration is not merely an algorithmic convenience—it computes the principal eigenvector of the Google Matrix $\widetilde{M} = dM_{2N}^T + (1-d)E_{2N}\mathbf{1}^T$. Eigendecomposition of $\widetilde{M}$ directly would require ~ 300 GB of memory; power iteration requires ~ 50 MB.
5. **Simultaneous dual improvement**: The model achieves a **95.7%** reduction in Top-100 MRE and a **76.2%** reduction in Overall MRE simultaneously, confirming the fix is physically principled and not over-fitted.

Empirical Result

By injecting physical reality into the mathematics—traversal time, phase transitions, spatial smoothing, and eigenvector theory—the algorithm transforms an inaccurate network-flow model into a state-of-the-art traffic prediction system without any machine-learning machinery.